

DecodeLabs

E-Commerce Sales & Customer Analytics

1.0 Problem Statement

E-commerce businesses collect large amounts of order data, but extracting meaningful insights from this information can be challenging. This project analyzes transactional data to uncover trends in sales performance, customer behavior, product demand, payment preferences, coupon usage, and referral effectiveness. The goal is to provide actionable insights that help improve business performance and support strategic decision-making.

2.0 Methodology

The project followed a structured analytics workflow consisting of two major phases: **Data Cleaning & Validation** and **Exploratory Data Analysis (EDA)**. SQL Server was used to assess data quality, validate business rules, and generate analytical insights from the e-commerce orders dataset.

2.1 Data Cleaning & Validation

The dataset was reviewed to ensure accuracy, consistency, and reliability before analysis. The following validation checks were performed:

- Verified the total number of unique orders and customers.
- Assessed the dataset date range to understand the reporting period.
- Checked for duplicate `order_id` values.
- Checked for missing values across all key fields.
- Validated business rules by comparing `total_price` against `unit_price × quantity`.
- Reviewed categorical fields (`product`, `payment_method`, `order_status`, `coupon_code`, and `referral_source`) for inconsistencies.
- Detected leading and trailing spaces in text fields such as customer IDs, shipping addresses, and tracking numbers.

2.2 Exploratory Data Analysis (EDA)

After validating the dataset, SQL queries were used to analyze business performance and customer behavior. Key analytical areas included:

Sales Performance Analysis

- Calculated total orders, revenue, quantity sold, and average unit price.
- Evaluated annual revenue trends.
- Measured Year-over-Year (YoY) revenue growth using window functions.

Customer Analysis

- Identified top-spending customers based on total purchase value.
- Assessed customer contribution to overall revenue.

Product Performance Analysis

- Analyzed order volume by product.
- Measured revenue generated by each product.
- Evaluated average quantity purchased per product.

Payment Method Analysis

- Determined the most frequently used payment methods.
- Compared revenue generated across payment channels.

Coupon Performance Analysis

- Evaluated coupon usage frequency.
- Measured revenue associated with each coupon code.

Referral Source Analysis

- Analyzed order acquisition by referral source.
- Assessed revenue contribution from different marketing channels.

2.2.1 Analytical Approach

To ensure business metrics reflected actual completed transactions, cancelled orders were excluded from revenue-based calculations and performance analyses. Aggregation functions, grouping operations, Common Table Expressions (CTEs), and window functions were used to generate meaningful business insights and trend analyses.

3.0 Findings

3.1 Data Quality Findings

A comprehensive data quality assessment was conducted to evaluate the completeness, consistency, accuracy, and validity of the dataset before analysis.

- The dataset covers a period from **January 1, 2023, to June 30, 2025**, providing approximately two and a half years of transactional data.
- No records were found with order dates beyond the maximum recorded date, indicating there were **no future-dated transactions**.
- Duplicate checks confirmed that **all 1,200 orders had unique Order IDs**, with no duplicate records detected.
- Missing value assessments across all key fields returned no results, indicating **100% completeness** for the analyzed attributes.
- Validation of the business rule $\text{Total Price} = \text{Unit Price} \times \text{Quantity}$ identified no inconsistencies, confirming that revenue calculations were accurate.
- Categorical field reviews showed standardized values across products, payment methods, order statuses, coupon codes, and referral sources, with no unexpected categories detected.
- Date validation revealed that transactions were distributed across three years:
 - 2023: 510 orders
 - 2024: 459 orders
 - 2025: 231 orders
- Text formatting checks found no leading or trailing spaces in critical fields such as Order ID, Customer ID, Shipping Address, and Tracking Number.

Overall, the dataset demonstrated a high level of quality and required no corrective cleaning before analysis.

3.2 Business Performance Findings

Order and Customer Overview

- The dataset contains **1,200 unique orders** generated by **1,189 unique customers**, indicating that most customers placed only one order during the analysis period.
- Excluding cancelled orders, the business processed **950 completed transactions**.

Order Status Analysis

Order Statuses were relatively evenly distributed:

Status	Orders
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

The high number of cancelled and returned orders suggests an opportunity to investigate fulfillment processes, product quality, or customer experience issues.

Revenue Performance

For non-cancelled orders:

- Total Revenue: **\$988,365.75**
- Total Quantity Sold: **2,776 units**
- Average Unit Price: **\$356.32**
- Highest Order Value: **\$3,456.40**
- Lowest Order Value: **\$14.06**

Annual revenue trends showed declining sales performance over time:

Year	Revenue
2023	\$551,643.24
2024	\$480,235.87
2025	\$231,882.85

Year-over-Year analysis further confirmed this decline:

- Revenue decreased by **12.74%** from 2023 to 2024.
- Revenue decreased by **56.65%** from 2024 to 2025.

This downward trend may indicate reduced customer demand, fewer transactions, seasonal limitations in the dataset, or changing business conditions.

Customer Analysis

The highest-spending customer generated **\$3,456.40** in total purchases, while the remaining top customers spent between approximately **\$3,294 and \$3,391**.

The relatively small difference between the top ten customers suggests that revenue is not heavily dependent on a single customer and is distributed across a broader customer base.

Product Performance Analysis

Order Volume

The most frequently ordered products were:

Product	Orders
Printer	146
Tablet	145
Laptop	138
Desk	135
Chair	133
Monitor	128
Phone	125

Revenue Contribution

Product	Orders
Printer	\$156,937.45
Tablet	\$148,364.84
Laptop	\$146,959.13
Desk	\$145,938.99
Chair	\$138,593.97
Monitor	\$127,872.24
Phone	\$123,699.13

Key Observations:

- **Printers** generated both the highest order volume and highest revenue.
- **Laptops** and **Chairs** also contributed significantly to overall sales.
- **Phones** generated the lowest revenue among the product categories analyzed.

Average quantity purchased per order was generally between two and three units, with Chairs and Laptops averaging three units per transaction.

Payment Method Analysis

Order Volume

Payment Method	Orders
Online	205
Cash	197
Debit Card	188
Gift Card	180
Credit Card	180

Revenue Contribution

Payment Method	Orders
Online	\$206,263.57
Cash	\$203,885.58
Debit Card	\$199,762.26
Gift Card	\$194,340.63
Credit Card	\$184,113.71

Online payments generated the highest order volume and revenue, suggesting that customers show a preference for digital payment channels.

Coupon Performance Analysis

Order Volume

Coupon Code	Orders
No Coupon	251
FREESHIP	246
SAVE 10	228
WINTER15	225

Revenue Contribution

Coupon Code	Revenue
No Coupon	\$261,335.60
FREESHIP	\$258,670.90
SAVE 10	\$245,893.08
WINTER15	\$222,466.17

Customers who did not use coupons generated the highest revenue, although the FREESHIP promotion performed nearly as well, suggesting that free shipping incentives may be more effective than discount-based promotions.

Referral Source Analysis

Order Volume

Referral Source	Orders
Instagram	218
Email	191
Facebook	186
Google	183
Referral	172

Revenue Contribution

Referral Source	Revenue
Instagram	\$224,619.38
Email	\$201,542.06
Facebook	\$199,475.16
Google	\$192,858.78
Referral	\$169,870.37

Instagram emerged as the strongest acquisition channel, generating both the highest number of orders and the highest revenue. Referral traffic produced the lowest revenue contribution, indicating that social and digital marketing channels were more effective at driving sales during the analysis period.

4.0 Recommendations

Recommendations

Based on the analysis of sales performance, customer behavior, product demand, payment preferences, promotional campaigns, and referral channel effectiveness, the following recommendations are proposed to support revenue growth, improve operational efficiency, and enhance customer acquisition strategies.

1. Improve Order Fulfillment and Customer Retention

The analysis revealed that cancelled and returned orders accounted for a significant portion of all transactions. While the dataset does not provide the specific reasons behind these outcomes, reducing cancellations and returns should be a key business priority.

The organization should investigate common causes of cancelled and returned orders by examining factors such as product availability, delivery delays, shipping accuracy, product quality, and customer expectations. Collecting and analyzing return reasons can help identify recurring issues and support targeted corrective actions.

Improving fulfillment processes and customer satisfaction can increase the percentage of completed transactions, ultimately contributing to higher realized revenue and stronger customer retention.

2. Expand High-Performing Product Categories

Product performance analysis identified Printers, Laptops, Chairs, and Tablets as the strongest contributors to revenue. These products consistently generated both high order volumes and substantial sales value.

The business should prioritize inventory planning, marketing efforts, and promotional campaigns around these high-performing product categories. Ensuring adequate stock availability and visibility for these products can help maximize revenue opportunities and reduce the risk of lost sales due to stock shortages.

Additionally, cross-selling and bundle promotions involving these top-performing products may encourage customers to increase their purchase value.

3. Optimize Underperforming Product Segments

Products such as Phones and Desks generated comparatively lower revenue despite maintaining moderate order volumes. This may indicate pricing challenges, lower customer demand, or increased market competition.

Management should conduct further product-level analysis to evaluate profit margins, customer preferences, and pricing competitiveness. Targeted promotions, product repositioning, or revised pricing strategies may help improve the performance of these categories.

4. Strengthen Digital Payment Adoption

Online payments generated the highest number of orders and the highest revenue contribution among all payment methods. This indicates strong customer preference for digital payment channels.

The organization should continue investing in seamless online payment experiences by optimizing checkout processes, improving transaction security, and reducing payment friction. Incentives such as loyalty points, cashback offers, or exclusive online payment discounts may further encourage adoption and improve conversion rates.

Maintaining multiple payment options remains important to accommodate different customer preferences and minimize abandoned purchases.

5. Refine Promotional and Coupon Strategies

Coupon analysis showed that customers who made purchases without using coupons generated the highest overall revenue. However, the FREESHIP promotion performed nearly as well, suggesting that free shipping incentives may be particularly effective in encouraging purchases.

Rather than relying solely on discount-based promotions, the business should experiment with shipping incentives, targeted promotions, and personalized offers based on customer behavior. Future campaign evaluations should measure not only order volume but also profitability and customer lifetime value.

Segmented promotions tailored to customer preferences may produce better results than broad, generic discount campaigns.

6. Increase Investment in High-Performing Marketing Channels

Referral source analysis identified Instagram as the strongest acquisition channel, generating both the highest order volume and the highest revenue contribution. Email and Facebook also demonstrated strong performance.

Given these results, marketing resources should be allocated strategically toward channels that generate the greatest return on investment. Increased investment in social media advertising, influencer partnerships, and audience engagement initiatives on Instagram may help sustain customer acquisition growth.

At the same time, lower-performing channels such as traditional referrals should be reviewed to determine whether process improvements or revised incentive structures could improve effectiveness.

7. Develop Customer Loyalty and Retention Programs

The customer analysis revealed that spending was relatively distributed across customers, with no single customer accounting for a disproportionate share of revenue. While this reduces dependency on individual buyers, it also highlights an opportunity to increase customer lifetime value.

The organization should consider implementing loyalty programs, personalized product recommendations, repeat-purchase incentives, and customer engagement campaigns designed to encourage repeat business.

Building stronger customer relationships can improve retention rates and increase long-term revenue generation.

8. Enhance Performance Monitoring and Reporting

The dataset demonstrated excellent data quality, with no duplicate records, missing values, formatting inconsistencies, or pricing calculation errors identified during validation. Maintaining these standards is critical for reliable decision-making.

The business should establish regular reporting dashboards that track key performance indicators such as revenue, order volume, cancellation rates, return rates, product performance, payment trends, and marketing channel effectiveness. Continuous monitoring will enable management to identify emerging trends and respond proactively to changes in customer behavior.

9. Interpret 2025 Revenue Trends with Caution

Although annual revenue appears lower in 2025 compared to previous years, this finding should not be interpreted as a definitive decline in business performance. The dataset only contains transactions through June 30, 2025, whereas the 2023 and 2024 figures represent complete calendar years.

For a more accurate comparison, future analyses should compare equivalent reporting periods (e.g., January–June 2024 versus January–June 2025) or wait until a full year of 2025 data becomes available. This will provide a more reliable assessment of revenue growth trends and business performance over time.